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CSC 34300

Lab 1: Control System for a 4-Floor Elevator

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Objective: We were tasked with developing a control system for a four-floor elevator using combinational logic implemented in VHDL. The control system is designed with eight inputs, consisting of four buttons for calling the elevator from the ground floor to the third floor (upward) and four buttons for calling it from the third floor to the ground floor (downward). The system has four outputs: Up, Down, F1, and F0. Up activates when any of the upward buttons are pressed, while Down functions the same for the downward buttons. The outputs F1 and F0 are two bits corresponding the requested floor, where 00 corresponds to the ground floor and 11 corresponds to the third floor. Several conditions must be met in the design: upward and downward movements cannot be requested at the same time, and if multiple buttons are pressed at once, the system prioritizes the highest requested floor. Furthermore, if the up and down buttons are pressed on the same floor, the system gives precedence to the upward movement.

Truth Table:

UP3	UP2	UP1	UP0	UP	DOWN	F1U	F0U
0	0	0	0	0	0	-	-
0	0	0	1	1	0	0	0
0	0	1	-	1	0	0	1
0	1	-	-	1	0	1	0
1	-	-	-	1	0	1	1

Down3	Down2	Down1	Down0	UP	DOWN	F1D	F0D
0	0	0	0	0	0	-	-
0	0	0	1	0	1	0	0
0	0	1	-	0	1	0	1
0	1	-	-	0	1	1	0
1	-	-	-	0	1	1	1

Boolean Equation:

From truth table

UP= UP0 or UP1 or UP2 or UP3
DOWN= DOWN0 or DOWN1 or DOWN2 or DOWN3
F0U=UP1 or UP3 **(WRONG WILL EXPLAIN LATER)**
F1U=UP2 or UP3
F0D= DOWN1 or DOWN3 **(WRONG WILL EXPLAIN LATER)**
F1U=DOWN2 or DOWN3

From VHDL

F0U=(UP1 and not UP2)or UP3 **(IF Up2 and Up1 are press it should be 1,0(F1,F0) but without** and not UP2 **it would be 1,1 and that would be the wrong floor)**

F0D= (DOWN1 and not DOWN2) or DOWN3 (**Same explanation as above**)

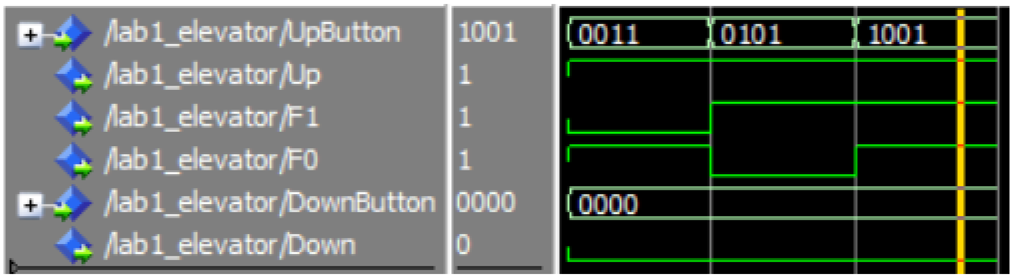
DOWN= (DOWN0 or DOWN1 or DOWN2 or DOWN3) and (not UP) (**Added** not UP **for the condition change where up and down can not be on at the same time and if it is then only Up will be on)**
F1 = F1U or F1D

F0= F0u or (F0d and not (F1u or F1d))

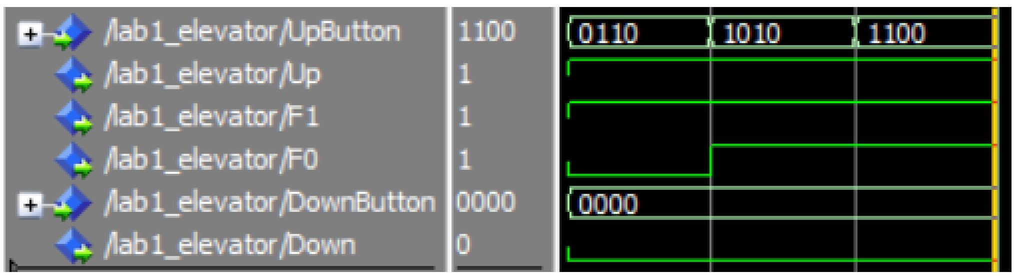
ModelSim Screenshots:

UP All Floor

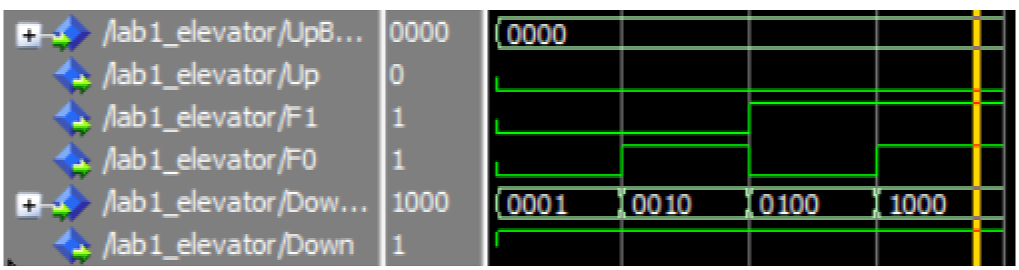
UP Floor -,0



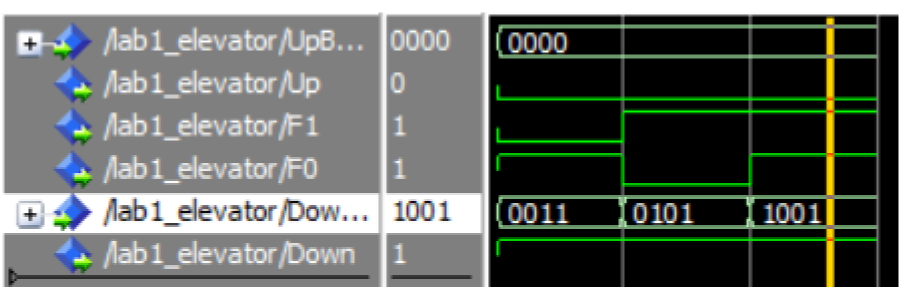
UP Floor -,1 and 3,2



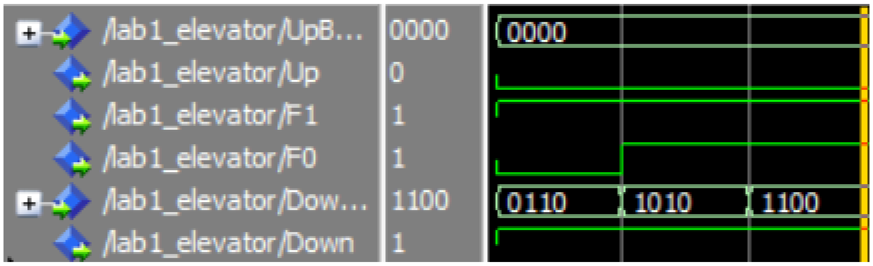
DOWN All Floors



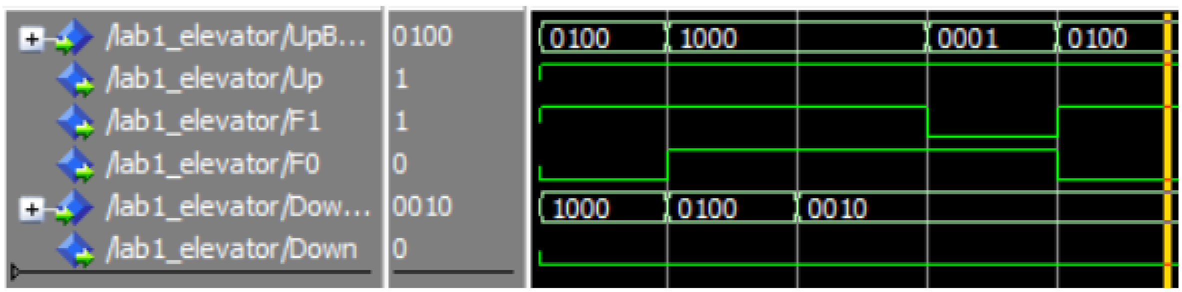
Down Floor -,0



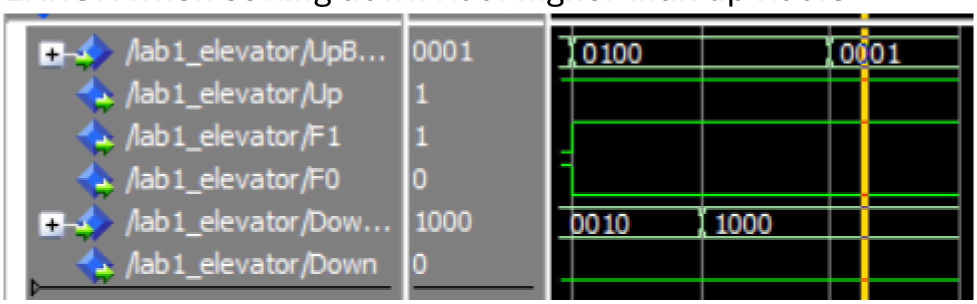
Down Floor -,1 and 3,2



Different up and down floors where up is higher than down



ERROR when setting down floor higher than up floors



Signal in design:

I added 5 signal 4 being for F1 and f0 for each up and down to make the final output easier to read and I added a signal of UpBoth where it's just the Up output, but I did not want to use a buffer for Up to make it work with the Down output expression.

Train of thought:

I was quite confused of how to implement the multiple buttons press for the truth table but after I realize I could just use don't care then it wasn't a problem any more. After that I implemented the truth table to VHDL and used ModelSim to test different cases for example if I were to press Up1 and Up2 the floor should be floor two but an error would occur where it would try to go to floor 3 instead so to fix this I add more condition to F0 which was F0U = (UP1 and not UP2)or UP3. The not UP2 makes it so that if UP2 is on with up1 it would force F0u to be 0 and keep it 1,0 and not 1,1. This other process would have similar problems and were need of fixing.

Some challenges I faced were the standard syntax error, having to wait every time for compiling and one error I have yet to find a solution for is that when testing the F0 and F1 where if up2 is press and down0 or down1 is pressed, then press down4 (F1,F0) would be 1,0 instead of 1,1 even after that if I would try to press Up0 instead the result will be the same 1,0. I believe the reason why this happens is due to my F0 output where I give try to give priority to high floor but have yet to find a solution for it.